

Risk Across Tails and Timescales

Does tail dependence change with the investment horizon,
and what does ignoring it do to measured risk?

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Round 1 - Initial Screening Challenge

Summary

We study a five-asset portfolio (US equities, Nasdaq, long Treasuries, gold and high-yield credit) on daily data from 12 April 2007 to 30 June 2026 (4,835 trading days). We use the maximal overlap discrete wavelet transform (MODWT) to look at the same returns at horizons from one day to more than three months, measure how often assets fall into their worst 5% together at each horizon, and then test what this does to Value-at-Risk (VaR) in a rolling out-of-sample backtest.

- **Does tail dependence change with the horizon?** Strong evidence for equities and credit, suggestive for equities and Treasuries, not detectable for gold or for the portfolio as a whole. Equities and high-yield credit crash together more as the horizon grows: their lower-tail co-exceedance rises from 0.53 at one day to 0.77 beyond 64 days, and the 90% interval for the change (+0.10 to +0.37) excludes zero (which holds across 63-, 126- and 252-day block lengths). For equities and Treasuries, co-exceedance rises from 0.05 to 0.22 (interval includes zero), while for gold it changes from 0.14 to 0.07 (interval also includes zero). For the portfolio’s risky sleeve against its hedge sleeve, the change is not statistically significant at any horizon.
- **What does ignoring it do?** A copula fitted to the matching horizon would give 5.0% higher 20-day VaR on average than a daily copula (and more than 10.9% higher on one day in ten). This compares two model forecasts on the same dates; out-of-sample evidence that the higher number is better is weak, as the Diebold-Mariano test cannot statistically separate them ($p = 0.32$ at 20 days, and at 5 days the daily $\sqrt{h}$ model actually scores best). The common $\sqrt{h}$ rule hides this, because it overstates 20-day VaR by even more (8.7% on average).
- **Recommendation.** For positions held 20 trading days or longer, stop scaling 1-day VaR by $\sqrt{h}$. Use the GARCH volatility forecast for the full horizon and a copula fitted to the matching wavelet horizon. Out of sample this model achieved the best point-estimate FZ score at 20 days (though without statistical significance) and needed 4.7% less capital on average than the $\sqrt{h}$ rule, while correct coverage (2 breaches in 191 non-overlapping windows) has low power to discriminate models (Section 4).

1 Question, data and portfolio

Risk models are usually fitted on daily returns, but many portfolios are judged over weeks or months. If the way assets crash together depends on the horizon, a daily model gets the long-horizon tail wrong even when its volatility forecasts are right. We ask whether that happens and how much it matters for measured risk.

We use five liquid ETFs that each stand for a different source of risk, with fixed weights typical of a balanced portfolio: SPY (S&P 500, 30%), QQQ (Nasdaq 100, 20%), TLT (20+ year Treasuries, 25%), GLD (gold, 15%) and HYG (high-yield credit, 10%). We call SPY, QQQ and HYG the *risky sleeve* and TLT and GLD the *hedge sleeve*. A two-sleeve view turns the question into one a portfolio manager cares about: does the hedge still pay off in a crash when you hold for longer?

Prices are daily adjusted closes. The sample starts on the first day HYG traded, so it includes the 2008 crisis, the 2011 downgrade, COVID in 2020, the 2022 rates shock and the 2025 tariff selloff. We use log returns. The portfolio return is calculated as the fixed-weight linear combination of constituent log returns ($r_{p,t} = \sum_i w_i r_{i,t}$), serving as a standard linear tracking proxy rather than an exact continuous-rebalancing geometric return. The prices are stored in the repository so the analysis runs without a download.

2 Method

2.1 Horizon views from the wavelet transform

The MODWT splits each return series r_t into detail series $D_1, \dots, D_J$ and a smooth S_J that add back to the original exactly, $r_t = \sum_{j=1}^J D_{j,t} + S_{J,t}$ [8]. Detail D_j holds movements that last roughly 2^j to 2^{j+1} days. We use $J = 5$, which reaches beyond 64 trading days (about a quarter), and the

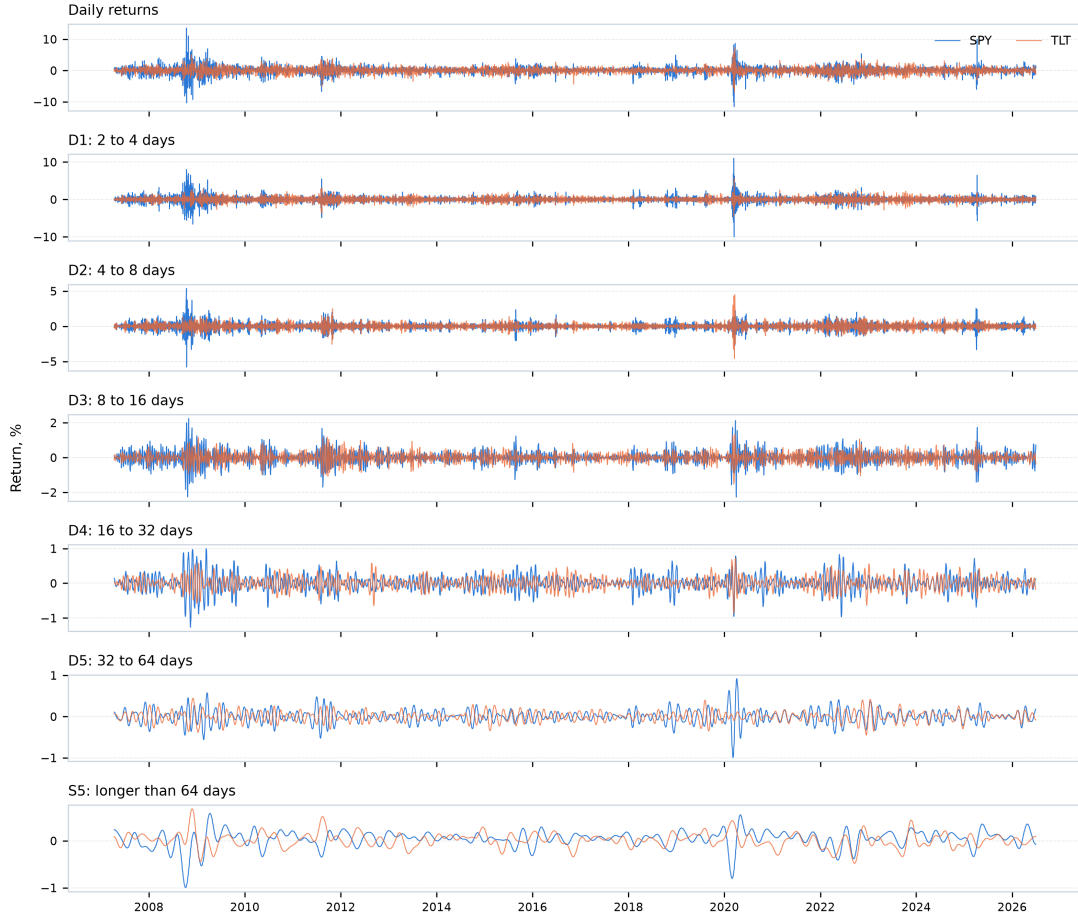


Figure 1: MODWT multiresolution analysis of SPY and TLT daily returns. The details and the smooth add back to the returns to machine precision.

sym8 filter. Figure 1 shows the result for SPY and TLT.

Detail bands on their own are not holding-period returns. A wavelet low-pass/MRA view is not identical to a literal h -day aggregated return; rather, we treat wavelet views as scale-matched proxies for investment horizons. An investor who holds for h days sees cumulative movements lasting longer than about h days, not a single narrow frequency band. So we build cumulative *horizon views*:

$$V_t^{(0)} = r_t, \quad V_t^{(j)} = \sum_{k=j+1}^J D_{k,t} + S_{J,t}, \quad j = 1, \dots, J, \quad (1)$$

where view j keeps movements longer than 2^{j+1} days. Because temporal aggregation behaves as a low-pass filter on daily returns, view j acts as a scale-matched proxy for an investment horizon of about 2^{j+1} days. Unlike non-overlapping h -day returns, it preserves every daily observation, avoiding sample truncation at coarse scales where non-overlapping returns leave fewer than 75 points.

Most of the variance sits at short scales: on average 79.6% of each asset’s variance is in D_1 and D_2 and only 5.2% in D_5 and S_5 (Table 1, reporting normalized variance contributions based on component sample variances). A copula fitted to daily returns is therefore dominated by what happens over a few days.

2.2 Measuring tail dependence at each horizon

For each view we convert every series to ranks, $u_{i,t} = \text{rank}(x_{i,t})/(n+1)$. Ranks remove the margins entirely, which is what a copula needs, and they are exactly reproducible. Our main measure is the

Table 1: Share of each asset’s return variance in each MODWT scale (%).

	SPY	QQQ	TLT	GLD	HYG
D1	62.1	60.7	56.3	55.2	54.5
D2	19.5	20.2	24.1	22.6	22.9
D3	9.9	10.2	10.5	11.5	11.9
D4	3.9	4.1	4.1	5.3	4.4
D5	2.0	2.1	2.0	2.6	3.0
S5	2.5	2.7	3.0	2.8	3.4

lower-tail co-exceedance at level q :

$$\hat{\lambda}_L(q) = \frac{1}{q} \Pr(U_1 \leq q, U_2 \leq q), \quad (2)$$

the share of one asset’s worst- q days on which the other is also in its worst q . It equals q for independent assets and 1 for perfectly dependent ones. We use $q = 5\%$, which leaves about 240 tail days at each horizon. The upper tail is defined the same way.

Correlation alone produces some co-exceedance, so we compare each estimate with what a Gaussian copula with the same normal-scores correlation ρ would give, $\Phi_2(\Phi^{-1}(q), \Phi^{-1}(q); \rho)/q$. Co-exceedance well above that line is tail dependence that correlation does not explain.

The smooth views are strongly autocorrelated, so they contain far fewer independent observations than their length suggests. We use a moving block bootstrap with blocks of 126 days (about six months, long enough to hold a few cycles of the slowest view) and 500 replications. The same resampled days are used for every view, so we can also put an interval on the change from daily to each horizon.

2.3 Copula family

For every pair and every view we fit Gaussian, Student- t , Clayton, Gumbel and Frank copulas by maximum likelihood and compare them by BIC. The t copula wins 40 of 42 contests, so all VaR models below use a t copula. Because it has symmetric tails, its job in the VaR model is to carry the overall strength of tail dependence at each horizon. The asymmetry between lower and upper tails is reported from the empirical co-exceedances.

2.4 VaR models

All copula models share the same margins, so any difference between them comes only from the dependence structure. In the multi-day rolling VaR implementation, each asset gets a constant-mean GJR-GARCH(1,1) model with Student- t innovations [4] (avoiding spurious drift accumulation over multi-day horizons, whereas the static marginal exploratory analysis uses AR(1)-GJR-GARCH(1,1)). The standardised residuals are modelled with an empirical body and generalised Pareto tails beyond the 10% and 90% points [6] (sensitivity checks across 5%, 10%, and 15% POT thresholds show that key tail-risk conclusions remain stable). For an h -day horizon, the variance is the sum of the GARCH forecasts for days $1, \dots, h$, so volatility mean reversion is included.

Our multi-day VaR construction generates an approximate horizon distribution by combining the cumulative h -day GARCH volatility forecast with a horizon-specific copula dependence structure; it is an analytical horizon approximation rather than a full multi-period path simulation that evolves daily GARCH volatility states sequentially over each step of the horizon.

We compare five models of the h -day portfolio loss at 99%:

- **Historical simulation** on overlapping h -day returns, and a **Gaussian** $\sqrt{h}$ rule: the two simple benchmarks.
- **Daily copula**, $\sqrt{h}$: 1-day VaR from the GARCH-EVT- t -copula model, scaled by $\sqrt{h}$. This is common practice.
- **Daily copula**, h -day **vol**: the same daily copula, with the full-horizon GARCH volatility instead

Table 2: Main modelling choices and why we made them.

Choice	Reason
MODWT, not DWT	Keeps all observations at every scale and is shift invariant, so results do not depend on the start date.
sym8, level 5	Near-symmetric filter with little phase distortion; level 5 reaches past a quarter while leaving enough data at the coarsest view.
Reflection boundary	Avoids wrapping the end of the sample onto its start, which matters in the rolling windows where the last day is the one that counts.
Cumulative horizon views	An investor holding h days sees all slower movements, not one band.
Ranks, not GARCH-filtered views	GARCH is badly behaved on smooth series and gave results that changed with library versions; ranks are exact and reproducible.
$q = 5\%$ co-exceedance	Enough tail days for stable estimates at every view; still in the tail.
Block bootstrap, 126 days	Keeps the autocorrelation of the smooth views; one block length for all views makes the change versus daily testable.
GJR-GARCH- t + EVT margins	Standard for daily risk; leverage effect and fat tails matter at 99%.
Summed GARCH variance	Allows volatility to revert to its mean over the horizon, which $\sqrt{h}$ ignores.
t copula	Wins 40 of 42 BIC contests.
Rolling 1,000-day window	About four years: long enough for GARCH and EVT, short enough to adapt.
FZ0 score and DM test	Coverage tests have little power at 20 days; FZ0 uses every forecast and scores VaR and ES together.

of $\sqrt{h}$.

- **Horizon copula (proposed):** the same margins, with the t copula fitted to the matching horizon view: view 1 (> 4 days) for $h = 5$, view 3 (> 16 days) for $h = 20$ and view 5 (> 64 days) for $h = 60$.

The second and third copula models differ only in volatility, the third and fourth only in dependence, so the comparisons separate the two effects. VaR and expected shortfall (ES) come from 20,000 simulated scenarios. Every model uses the same random numbers, so differences are not simulation noise.

2.5 Out-of-sample backtest

At the close of each day t we forecast the loss over days $t + 1$ to $t + h$ using only data up to t . All models are re-estimated every 21 trading days on the last 1,000 days, and the GARCH volatilities are updated every day in between. The wavelet views are computed inside each window, so the two-sided filter never sees the future. The first forecast is in March 2011, giving 3,835 forecast days per horizon.

We judge the models with the Kupiec coverage test [5], the Christoffersen independence test [2], the worst Basel-style traffic-light zone over any 250-day window (applied strictly as a diagnostic for the 1-day 99% VaR framework [1]; we do not imply this regulatory classification extends directly to multi-day horizons), and the Fissler-Ziegel FZ0 score for VaR and ES jointly [3, 7]. FZ0 is a strictly consistent scoring rule, so lower is better and it rewards a model that is neither too high nor too low. Overlapping h -day losses are serially dependent, so coverage tests use every h -th forecast, and FZ differences are tested with a Diebold-Mariano test using Newey-West standard errors.

Table 2 collects the main choices and the reason for each.

3 Results

3.1 Tail dependence across horizons

Table 3 and Figure 2 show the main result. Three things stand out.

Equity and credit crash together more at longer horizons. SPY-HYG lower-tail co-exceedance is 0.53 at one day and 0.77 beyond 64 days, well above the 0.44 a Gaussian copula with the same correlation would give. The change, +0.24 with a 90% interval of +0.10 to +0.37, is the clearest horizon effect in the data and remains robustly positive across 63-, 126-, and 252-day block lengths. Credit losses build over weeks as spreads widen, so a credit position that looks only partly linked to equities day by day is closely linked over a quarter.

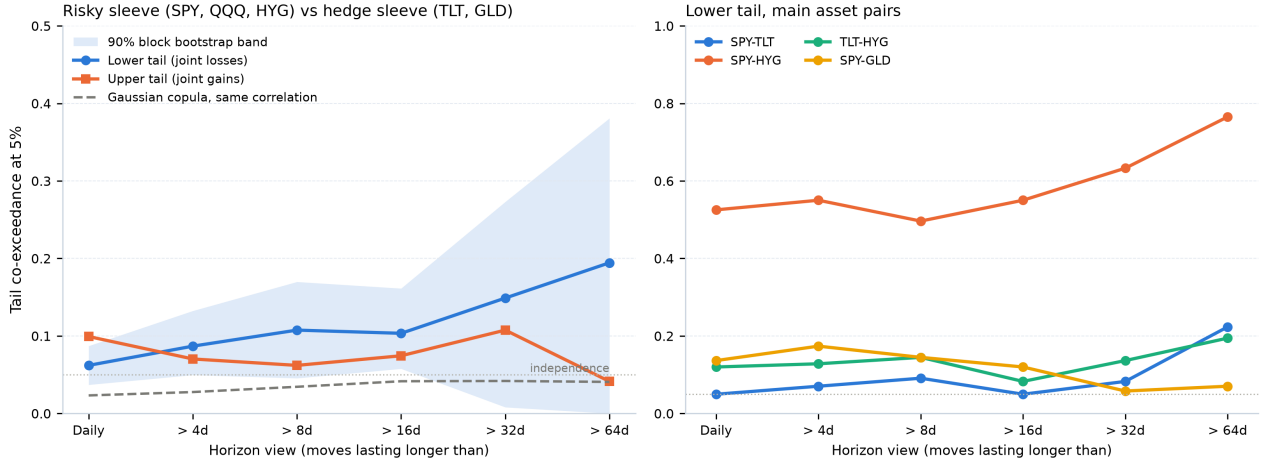


Figure 2: Left: lower and upper tail co-exceedance of the risky and hedge sleeves by horizon view, with the 90% block bootstrap band and the Gaussian copula benchmark. Right: lower-tail co-exceedance for the main asset pairs. The dotted line is independence.

Suggestive evidence for Treasuries; gold shows no detectable horizon shift. On daily data SPY and TLT almost never crash together (0.05, about what independence would give). Beyond 64 days the figure is 0.22 (interval 0.01 to 0.34, which includes zero). All of the joint crash days at that horizon come from two episodes, early 2009 and 2022, when stocks and long bonds fell together for months. For gold, co-exceedance shifts from 0.14 to 0.07, but its bootstrap interval spans zero (-0.12 to $+0.03$), so no horizon shift is statistically detectable.

Portfolio sleeve-level test is not statistically significant. Co-exceedance of the risky and hedge sleeves is 0.06 at one day and 0.19 beyond 64 days. At every horizon it sits above the Gaussian benchmark (0.02 at one day). However, the change of $+0.13$ has a 90% bootstrap interval of -0.05 to $+0.33$, which includes zero at all horizons: the longest views contain few independent episodes, and the portfolio-level test is not statistically significant. The upper tail does not rise in the same way, so any co-movement is concentrated in crashes rather than rallies.

Table 3: Lower-tail co-exceedance at the 5% level by horizon view. Brackets are 90% moving block bootstrap intervals.

Horizon	Risky vs hedge	Change vs daily	Gaussian	SPY-TLT	SPY-HYG	SPY-GLD
Daily	0.06 [0.04, 0.09]	—	0.02	0.05	0.53	0.14
> 4d	0.09 [0.05, 0.13]	+0.02 [+0.00, +0.07]	0.03	0.07	0.55	0.17
> 8d	0.11 [0.05, 0.17]	+0.05 [+0.00, +0.11]	0.03	0.09	0.50	0.14
> 16d	0.10 [0.06, 0.16]	+0.04 [+0.00, +0.10]	0.04	0.05	0.55	0.12
> 32d	0.15 [0.01, 0.27]	+0.09 [-0.05, +0.22]	0.04	0.08	0.63	0.06
> 64d	0.19 [0.00, 0.38]	+0.13 [-0.05, +0.33]	0.04	0.22	0.77	0.07

3.2 What ignoring it does to measured risk

Figure 3 compares the horizon copula with the daily copula over the whole backtest. Both use the same margins and the same full-horizon volatility, so the gap is purely the effect of the dependence structure (stemming from both correlation shifts and the copula tail parameter). A copula fitted to the matching horizon would give 2.1% higher VaR at 5 days, 5.0% higher at 20 days and 4.3% higher at 60 days on average than a daily copula. The effect varies over time: at 20 days the 10th to 90th percentile range is -1.0% to $+10.9\%$, and at 60 days -10.7% to $+15.9\%$, because the coarsest view is estimated on few independent episodes.

The usual $\sqrt{h}$ rule makes the daily copula look safe. Scaling 1-day VaR by $\sqrt{h}$ gives 8.7% more VaR at 20 days than the full-horizon GARCH volatility and 18.2% more at 60 days, because high volatility tends to fall back over the horizon. That overstatement is larger than the dependence understatement, so the $\sqrt{h}$ rule ends up with enough capital for the wrong reasons. A desk that

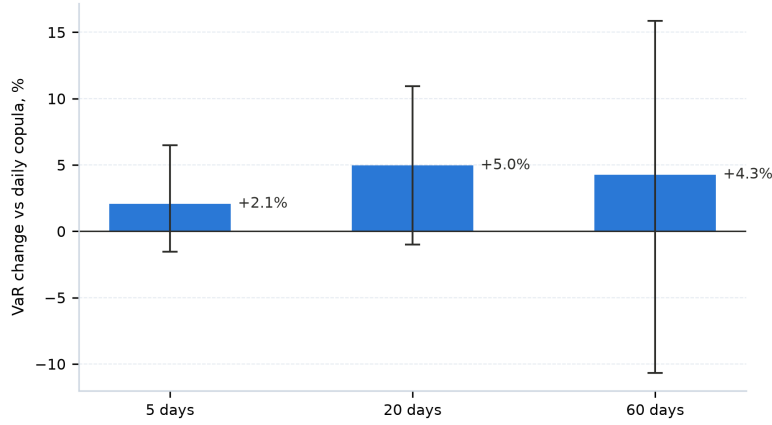


Figure 3: Average change in VaR from using the horizon copula instead of the daily copula, with the same margins and volatility. Bars are means over all forecast days; whiskers are the 10th and 90th percentiles.

improves its volatility model but keeps daily dependence would remove the cushion and be left with too little capital.

3.3 Out-of-sample backtest

Table 4 reports the backtest. At one day the three copula models are the same forecast. They give 37 breaches against 38 expected, pass both coverage tests (Kupiec $p = 0.83$, Christoffersen $p = 0.40$) and reach the yellow zone at worst, in the 250 days after the COVID crash. Both simple benchmarks fail: the Gaussian rule has 68 breaches and historical simulation, while close on the total, bunches its 41 breaches together; both reach the red zone.

At 20 days the horizon copula achieved the best point-estimate FZ score of all five models, with 2 breaches in 191 non-overlapping windows against 1.9 expected. However, this point-estimate improvement is not statistically significant ($p = 0.32$ against the daily copula, $p = 0.25$ against $\sqrt{h}$), and correct coverage resting on 2 breaches in 191 windows has weak power to reject misspecification. At 5 days the daily $\sqrt{h}$ model actually scores best, with all copula models within noise of each other ($p = 0.51$).

Every model failed in 2022. During the 2022 rates shock, stocks and bonds fell together for months. Over its 209 forecast days at 20 days, all copula models experienced severe cluster breaches against approximately 2 expected: the daily copula had 16 breaches and the horizon copula 13 (a modest gain of only 3 overlapping days). The $\sqrt{h}$ rule recorded 11 breaches, but only because it carried excessive capital throughout the sample. In the COVID crash all three models recorded 5 breaches: the shock was too abrupt for any 20-day model. Overlapping multi-day forecasts are serially dependent, so these stress-period counts serve as descriptive diagnostics rather than formal evidence of model superiority.

4 Recommendation

For positions held 20 trading days or longer, replace the $\sqrt{h}$ rule with a full-horizon GARCH volatility forecast and a copula fitted to the matching wavelet horizon view.

Why. On this portfolio the change needs 4.7% less 20-day capital on average than $\sqrt{h}$ (6.8% against 7.1% of portfolio value), achieves the best point-estimate FZ score among the five models (though not statistically significant), and aligns model structure with the holding horizon. The $\sqrt{h}$ rule overstates 20-day VaR on average and understates long-horizon dependence, with the first error masking the second on average. That does not hold on every day: when volatility is low the $\sqrt{h}$ rule understates it too, and both errors point the same way. Nor does it hold during severe regime shifts where every model failed, as observed in 2022.

How to do it tomorrow. The change fits on top of an existing GARCH-copula VaR system:

Table 4: Rolling out-of-sample backtest of 99% VaR and ES. Coverage tests use non-overlapping windows. FZ is the Fissler-Ziegel score (lower is better); ΔFZ is the difference to the daily copula with its Diebold-Mariano p -value.

h	Model	Breaches / exp.	Kupiec p	Christ. p	FZ	ΔFZ (p)	Avg VaR	Basel
1d	Historical simulation	41 / 38.4	0.67	0.00	-3.612	+0.308 (0.05)	1.75%	Red
1d	Gaussian, $\sqrt{h}$	68 / 38.4	0.00	0.00	-3.381	+0.539 (0.06)	1.47%	Red
1d	All three copula models	37 / 38.4	0.83	0.40	-3.920	—	1.60%	Yellow
5d	Historical simulation	6 / 7.7	0.53	0.76	-2.686	+0.271 (0.23)	3.92%	—
5d	Gaussian, $\sqrt{h}$	10 / 7.7	0.42	0.11	-2.379	+0.578 (0.17)	3.19%	—
5d	Daily copula, $\sqrt{h}$	10 / 7.7	0.42	0.61	-2.973	-0.015 (0.19)	3.58%	—
5d	Daily copula, h -day vol	10 / 7.7	0.42	0.61	-2.957	—	3.45%	—
5d	Horizon copula (proposed)	10 / 7.7	0.42	0.61	-2.966	-0.008 (0.51)	3.52%	—
20d	Historical simulation	3 / 1.9	0.46	0.76	-1.804	+0.432 (0.18)	7.33%	—
20d	Gaussian, $\sqrt{h}$	2 / 1.9	0.95	0.84	-1.861	+0.376 (0.27)	6.03%	—
20d	Daily copula, $\sqrt{h}$	1 / 1.9	0.47	0.92	-2.233	+0.003 (0.95)	7.15%	—
20d	Daily copula, h -day vol	2 / 1.9	0.95	0.84	-2.237	—	6.49%	—
20d	Horizon copula (proposed)	2 / 1.9	0.95	0.84	-2.271	-0.034 (0.32)	6.81%	—

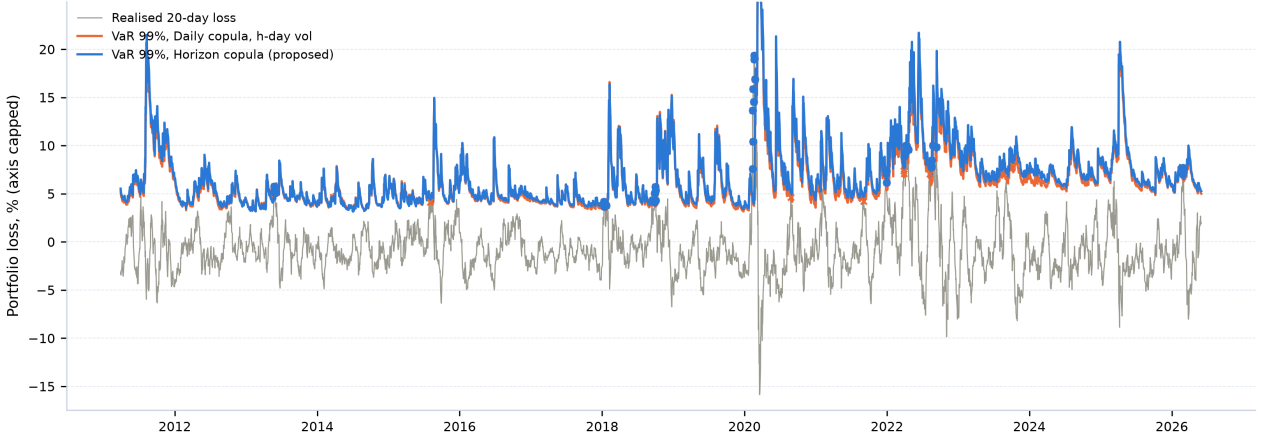


Figure 4: Realised 20-day portfolio loss against the 99% VaR of the daily copula and the horizon copula, March 2011 to June 2026. Markers show breaches. The axis is capped so the March 2020 peak does not flatten the rest.

1. Keep the daily GARCH-EVT margins. For the h -day horizon, sum the GARCH variance forecasts for days $1, \dots, h$ instead of multiplying the 1-day variance by h .
2. Once a month, compute the MODWT of the last four years of returns, build the horizon view that matches the holding period (view 3, moves longer than 16 days, for a 20-day horizon) and refit the copula to its ranks.
3. Simulate h -day losses from the new copula and the h -day margins.

All three steps are in `src/horizon_var.py`. The monthly refit adds a few seconds of computing time.

What to watch. Report the horizon-matched VaR next to the daily-copula VaR. When the gap widens, as it did when peaking between 2021 and 2023 (9.7%, 7.9%, and 10.1% by year), the portfolio's hedges are working less well over the holding period than daily data suggests, which is a signal to review the hedge rather than only to hold more capital.

5 Limitations

- The longest views contain few independent crash episodes. This is why the sleeve-level change and the 60-day capital gap have wide intervals. More history would help, but HYG only starts in 2007.
- The backtest needs 1,000 days of history, so it begins in March 2011. The 2008 crisis is used for estimation only and is never forecast out of sample.
- Coverage tests at 20 days rest on 191 non-overlapping windows, so they cannot tell good models

apart. We rely on the FZ score, which uses every forecast, and still find differences that are not statistically significant ($p = 0.32$ at 20 days; at 5 days daily $\sqrt{h}$ scores best).

- The horizon copula is fitted to ranks of the smoothed returns without removing volatility clustering first. Shared volatility spikes may add to measured dependence. The daily copula is fitted to the same kind of data, so the comparison between them is fair, but the levels should be read with this in mind.
- The multi-day VaR models reuse the daily EVT marginal shock shape scaled by the h -day volatility forecast. Because temporal aggregation typically thins heavy tails toward normality, reusing daily marginal tail shapes may overstate tail heaviness at longer holding periods (such as 20 days).
- Our multi-day VaR construction produces an approximate horizon distribution combining horizon volatility and horizon dependence; it is an analytical approximation rather than a full multi-period path simulation evolving daily GARCH states sequentially.
- The measured VaR gap between the daily and horizon copulas is not purely due to tail dependence: decomposition shows it stems partly from the correlation matrix and partly from the t -copula tail parameter (degrees of freedom), with the tail component varying in sign across rolling estimation windows.
- The portfolio weights are fixed and there are no transaction costs, rebalancing or liquidity effects.

6 Conclusion

Evidence that tail dependence changes with the horizon is strong for equities and credit, suggestive for equities and Treasuries, but not detectable for gold or for the portfolio as a whole. A risk model that uses a copula fitted to the matching horizon would give about 5.0% higher 20-day VaR on average than a daily copula, and more than 10.9% higher on one day in ten (comparing daily model forecasts on identical dates, with out-of-sample evidence that the higher number is better being weak, $p = 0.32$). The $\sqrt{h}$ rule hides this by overstating 20-day VaR. Measuring horizon risk with full-horizon volatility and horizon-matched dependence structure addresses both issues, achieved the best point-estimate FZ score at 20 days, and requires less capital on average than the rule it replaces.

Reproducibility and use of AI tools

`python run_all.py` rebuilds every number, table and figure in this report from the stored price data in about six minutes, and gives identical output on repeated runs on a single machine (runs across different CPU architectures or BLAS libraries can differ in a final digit due to floating-point rounding). The numbers in the text are written by the same script to `tables/key_numbers.tex`. Package versions are pinned in `requirements.txt` (Python 3.12), and `pytest` runs the test suite.

AI assistants were used for coding help, debugging and drafting text. The team chose the question, data, methods and conclusions, checked every result against the code, and takes full responsibility for the submission. Details are in `docs/AI_DISCLOSURE.md`.

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